Singular weights and operator dependence in matrix-valued transport

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Abstract

We study several parameter limits of the matrix-valued unbalanced transport problem of Li and Zou. For convergent transport and reaction weights, an eventual relative comparison of their Gram matrices with factors tending to one gives convergence of extended minimum values. At finite limiting cost, the minimum is attained, near-minimizers admit optimal weak-star clusters, and the corresponding Frobenius variations vanish in the null directions of the limiting weights. Fixed eigenbases and vanishing reaction weights follow as special cases. Along fixed rays or under uniform divergence, large transport weights have a different limit: the reaction-channel infimum over the original finite-action class. For entrywise spatial divergence on a convex domain with a positive-definite reaction weight, the uniform-divergence limit admits a weighted Bures formula in terms of the total endpoint matrices and an explicit Radon-measure recovery bound. Rotating families and an alternating norm-divergent sequence show the limits of value continuity and norm-based divergence assumptions. A smooth, uniformly positive example also separates convergence in the full matrix-flux and Kantorovich–Bures models. Together, these results show that singular parameter limits can depend essentially on directional geometry and the transport operator, not only on the magnitude of the weights.

1 Introduction

The general matrix-valued transport problem of Li and Zou combines a constant-coefficient continuity operator with a weighted quadratic transport and reaction action [7]. It includes the Kantorovich–Bures model of Brenier and Vorotnikov [2] and the spatial matricial interpolation model of Chen, Georgiou and Tannenbaum [3]. The choice of operator can affect limiting costs and convergence behavior, as the examples below show.

The paragraph following Proposition 4.6 of [7] identifies vanishing reaction weights, large transport weights, and partial eigenvalue degeneration as further limiting regimes. We retain their signed-Radon support formulation and finite positive semidefinite endpoints, extending the measure-level parameter analysis to singular reaction weights as well.

The weights enter the action through their Gram matrices. Under an eventual relative comparison with factors tending to one, Theorem 4.1 gives stability of extended minimum values for simultaneously convergent weights. For admissible paths with uniformly bounded actions, bounded weights give coordinate-measure compactness; a small scalar shift of a continuous mass-dual field proves the joint lower bound; relative comparison supplies constant-path recovery. The comparison is a sufficient condition; fixed eigenbases give it automatically. The reaction-zero regime is a corollary, including an infinite balanced value.

For diverging transport weights, we preserve the finite-action class and prove a direct infimum limit. Theorems 5.2 and 5.3 treat fixed singular rays and uniform divergence in all directions separately. For entrywise divergence on a convex domain with a positive-definite reaction weight, spatial aggregation and a three-part hub construction identify the uniform-divergence limit explicitly (Theorem 6.2). The Bures formula, its Procrustes representation

[1], and weighted reaction congruence [7, Remark 3.11] are existing tools; related weighted Bures geometry is studied in [5].

The examples in Section 7 illustrate the scope of these results. Rotation can create a finite excess cost or divergence in a partially degenerating reaction problem, and a jump in a scalar transport problem. Divergence of the Frobenius norm does not imply convergence of minimum values. Even isotropic divergence need not produce a total-matrix Bures value for the Kantorovich–Bures operator. Finally, Section 8 gives a high-frequency family of smooth, uniformly positive densities that separates convergence in two concrete unit-weight models.

Scalar dynamic transport/reaction limits already appear in [4]. The arguments use established convex duality, measure compactness and finite-dimensional matrix geometry. The discrete convergence studied in [6] concerns a different limiting parameter, whereas the present work varies the transport and reaction weights of the continuous problem.

2 Setup and notation

Let n, k, d be positive integers, let Ω be the closure of a bounded smooth domain in $\mathbb{R}^d$, and put $Q = [0,1] \times \Omega$. The explicit hub construction below additionally assumes convexity. For each size m, $\mathbb{S}^m$ is the space of real symmetric matrices, $\mathbb{S}^m_+$ its positive-semidefinite (PSD) cone, and $\mathbb{S}^m_{++}$ its symmetric positive-definite (SPD) cone. We write I_m for the identity, $O(m) = \{U \in \mathbb{R}^{m\times m} : U^T U = I_m\}$ for the orthogonal group, and $X^{1/2}$ for the unique PSD square root of $X \succeq 0$. Identity subscripts may be omitted when the size is fixed by the formula. The order $A \succeq B$ means $A - B \succeq 0$, and $\operatorname{sym} A = (A + A^T)/2$. Every real matrix space carries the Frobenius pairing $\langle A, B\rangle_F = \operatorname{tr}(A^T B)$ and squared norm $\|A\|_F^2 = \sum_{i,j} A_{ij}^2$. We also write $A \cdot B$ for this pairing and use it componentwise with matrix-valued measures, summing over all matrix entries. The transport dimension k and matrix dimension n are independent.

Fix a linear first-order constant-coefficient operator $D^* : C^1(\Omega, \mathbb{S}^n) \to C(\Omega, \mathbb{R}^{n\times k})$ with $D^* I_n = 0$. Its zeroth-order part is allowed. We use all closed-domain C^1 tests, with derivatives extended continuously from the interior, including tests without compact spatial support.

The endpoints G_0, G_1 are finite PSD matrix-valued Radon measures. A path $\mu = (G, q, R)$ consists of finite signed coordinate measures on Q, with symmetric G, full $n \times k$ flux q, and full $n \times n$ reaction R. Admissibility means

$$\int_Q \partial_t \Phi \cdot dG + D^*\Phi \cdot dq + \Phi \cdot dR = \int_\Omega \Phi(1) \cdot dG_1 - \int_\Omega \Phi(0) \cdot dG_0 \quad (\Phi \in C^1(Q, \mathbb{S}^n)). \tag{1}$$

Write $\mathrm{CE}_{D^*}(G_0, G_1)$ for this admissible class, and abbreviate it to CE when the operator and endpoints are fixed. Finite action implies positivity of the mass (Lemma 3.1). When an explicit construction is given by spatial measure curves (G_t, q_t, R_t), the associated space–time triple is

$$(G, q, R) = (dt\, G_t, dt\, q_t, dt\, R_t).$$

For constructions with Lebesgue densities we write $G_t = M(t,\cdot)\, dx$, $q_t = F(t,\cdot)\, dx$, and $R_t = Z(t,\cdot)\, dx$; individual flux blocks are specified locally.

For $L \in \mathbb{S}^k_+$ and $H \in \mathbb{S}^n_+$, the dual variables range over

$$(A, B, C) \in \mathbb{S}^n \times \mathbb{R}^{n\times k} \times \mathbb{R}^{n\times n}.$$

Thus B and C are full matrices. Define

$$\mathcal{O}_{L,H} = \{(A, B, C) : A + \tfrac{1}{2} BLL^T B^T + \tfrac{1}{2} CHH^T C^T \preceq 0\}. \tag{2}$$

The local cost is its support function,

$$j_{L,H}(g,z,r) = \sup_{(A,B,C)\in\mathcal{O}_{L,H}} (A\cdot g + B\cdot z + C\cdot r).$$

Let ρ_μ be the sum of the total variations of all coordinates of μ, and let (g,z,r) be their Radon–Nikodym densities. Set

$$\begin{aligned} \mathcal{J}_{L,H}(\mu) &= \int_Q j_{L,H}(g,z,r)\, d\rho_\mu, \\ V_{D^*}(L,H;G_0,G_1) &= \inf_{\mu\in \mathrm{CE}_{D^*}(G_0,G_1)} \mathcal{J}_{L,H}(\mu). \end{aligned} \tag{3}$$

We write $V(L,H)$ with the operator and endpoints fixed. The later values $W(L,H)$ and b_L suppress the same data; endpoint-dependent shorthand for fixed operators and weights will be declared in Section 8. These take values in $[0,+\infty]$; the infimum of the empty finite-action class is $+\infty$. This is the original action normalization of [7]: V is the minimum action, with a factor $1/2$ in each quadratic channel. In that notation, $(L,H)=(\Lambda_1,\Lambda_2)$, $j=J_\Lambda$, $\mathcal{J}=J_{\Lambda,Q}$, and $V=\mathrm{WB}^2_\Lambda$. For $g\in\mathbb{S}^n_{++}$, $L\in\mathbb{S}^k_{++}$, and $H\in\mathbb{S}^n_{++}$, square completion gives

$$j_{L,H}(g,z,r) = \tfrac{1}{2}\operatorname{tr}\big((zL^{-1})^T g^{-1}(zL^{-1})\big) + \tfrac{1}{2}\operatorname{tr}\big((rH^{-1})^T g^{-1}(rH^{-1})\big).$$

Larger Gram matrices LL^T and HH^T lower the action: L,H act as square-root mobilities. At singular weights the support definition retains every range constraint. Our normalization gives $E_H = 2d_B^2$ for transformed pure-reaction endpoints in Lemma 6.1, and $d^2_{\mathrm{KB}} = 2V_{\mathrm{KB}}$ for the Kantorovich–Bures model in Section 8. The support-function description and its finite-dimensional quadratic interpretation are due to [7]. Our proofs use the support function directly, including at singular weights. The following elementary consequences will also justify the reference measures and rescalings used in the constructions.

Lemma 2.1 (Support scaling and reference measures)**.** *For symmetric g and compatible z,r, and for $a,c>0$, the support cost satisfies, in the extended nonnegative reals,*

$$j_{L,H}(ag,az,ar) = a\,j_{L,H}(g,z,r), \qquad j_{L,H}(0,0,0)=0, \tag{4}$$

$$j_{L,H}(g,cz,cr) = c^2 j_{L,H}(g,z,r), \qquad j_{L,H}(ag,z,r) = a^{-1} j_{L,H}(g,z,r). \tag{5}$$

The action in (3) *can be computed using any finite positive measure dominating all coordinate variations of μ.*

Proof. Positive homogeneity follows directly by multiplying the linear pairing in a support supremum by $a>0$; at the zero argument every pairing is zero. The map $(A,B,C)\mapsto(c^2A,cB,cC)$ is a bijection of $\mathcal{O}_{L,H}$, because it multiplies its defining matrix inequality by c^2 and its inverse uses $1/c$. Substitution in the support supremum proves the first identity in (5). For the second, use

$$j_{L,H}(ag,z,r) = a\,j_{L,H}(g,z/a,r/a) = a\,a^{-2} j_{L,H}(g,z,r).$$

Finally let λ be such a dominating measure and $h=d\rho_\mu/d\lambda$. On $\{h>0\}$, the Radon–Nikodym chain rule gives $d\mu/d\lambda = h(d\mu/d\rho_\mu)$, while all coordinate densities vanish on $\{h=0\}$. Homogeneity and $j_{L,H}(0,0,0)=0$ therefore give

$$\int_Q j_{L,H}(d\mu/d\lambda)\, d\lambda = \int_{\{h>0\}} h\, j_{L,H}(d\mu/d\rho_\mu)\, d\lambda = \mathcal{J}_{L,H}(\mu).$$

All integrands are nonnegative, so the equality remains valid for infinite action, without an integrability assumption on the cost. □

For any matrix channel M of μ, let $m = dM/d\rho_\mu$. For a fixed compatible rectangular matrix A,

$$\|MA\|_{\mathrm{TV},F} = \int_Q \|mA\|_F \, d\rho_\mu. \tag{6}$$

This is the Frobenius norm variation of the product measure. In particular the reaction product is RA, with no symmetrization. Weak-star convergence below means convergence of every signed coordinate against every continuous scalar test on the underlying compact space, Q for space–time measures and Ω for spatial measures.

3 Common estimates and variational tools

We use the continuous-dual framework and positivity criterion of [7, Lemma 3.3 and Appendix A]. The trace and variation estimates below develop [7, Lemma 3.12 and Proposition 3.13] for varying PSD weights and fixed matrix directions.

Lemma 3.1 (Continuous representation and positivity)**.** *For every symmetric-mass path,*

$$\mathcal{J}_{L,H}(\mu) = \sup_{(A,B,C)\in C(Q,\mathcal{O}_{L,H})} \left(\int A \cdot dG + \int B \cdot dq + \int C \cdot dR \right). \tag{7}$$

Finite action forces $g \succeq 0$ almost everywhere, hence $G(S) \succeq 0$ for every measurable S. The fixed-weight action is weak-star lower semicontinuous.

Proof. The set $\mathcal{O}_{L,H}$ contains zero and is closed because its defining matrix expression is continuous and the cone of negative semidefinite matrices is closed. To verify convexity, for $0 \le \theta \le 1$ write $B_\theta = \theta B_1 + (1-\theta)B_2$. For every $K \succeq 0$,

$$\begin{aligned} &\theta B_1 K B_1^T + (1-\theta) B_2 K B_2^T - B_\theta K B_\theta^T \\ &\qquad = \theta(1-\theta)(B_1 - B_2) K (B_1 - B_2)^T \succeq 0. \end{aligned}$$

Use $K = LL^T$ and the corresponding identity for C with $K = HH^T$. The constraint at a convex combination of two dual triples is therefore bounded above by the convex combination of their constraint matrices, which is negative semidefinite.

For every continuous admissible dual field, the definition of the support function gives $A \cdot g + B \cdot z + C \cdot r \le j_{L,H}(g,z,r)$ pointwise. The signed Radon–Nikodym pairing and integration give the upper bound in (7).

For the reverse inequality, following [7, Appendix A], write $v = (g,z,r)$ and embed the dual set in the full-coordinate Hilbert space

$$\begin{aligned} \mathcal{E} &= \mathbb{R}^{n\times n} \times \mathbb{R}^{n\times k} \times \mathbb{R}^{n\times n}, \\ \|(A,B,C)\|_2^2 &= \|A\|_F^2 + \|B\|_F^2 + \|C\|_F^2 . \end{aligned}$$

The mass field of every admissible dual point remains symmetric. The inner product is the sum of the three Frobenius pairings. Thus both off-diagonal entries of each symmetric matrix are counted in its norm and pairing, exactly as in the action. The preceding calculation establishes that the dual set is nonempty, closed and convex.

Finite measurable selectors. Every subset of a finite-dimensional Euclidean space is second countable and therefore separable. Choose a countable dense sequence $(u_\ell)_{\ell\ge 0}$ in $\mathcal{O}_{L,H}$, with $u_0 = 0$. Continuity of a linear pairing in its dual argument gives, for every v,

$$j_{L,H}(v) = \sup_{\ell\ge 0} u_\ell \cdot v.$$

Indeed, every fixed point of the dual set is a limit of sequence points, so its pairing is bounded by this countable supremum. This reasoning also applies when the supremum is infinite.

Choose finite Borel representatives of the Radon–Nikodym densities, altering a null set if necessary, and put $a_\ell = u_\ell \cdot v$ and $m_N = \max_{0\le\ell\le N} a_\ell$. These are measurable scalar functions. Define $s_0 = u_0$ recursively: keep s_N when $a_{N+1} \le s_N \cdot v$, and otherwise take $s_{N+1} = u_{N+1}$. The comparisons are measurable, so each s_N is a measurable selector whose values belong to the finite set $\{u_0, \dots, u_N\}$; in particular it is bounded. This rule retains the earlier choice in a tie. It takes values in the dual set and satisfies $s_N \cdot v = m_N$. Because $u_0 = 0$, the functions m_N are nonnegative and increase pointwise to $j_{L,H}(v)$.

Weighted continuous approximation. Let v_a range over all matrix entries of the three densities, set $w = \sum_a |v_a|$, and define $d\nu = w\, d\rho_\mu$. The coordinate densities are integrable, so ν is a finite Borel measure and hence a Radon measure on the compact metric space Q. Moreover $\|v\|_2 \le w$. We give the approximation argument in this particular weighted measure. For a Borel set E and $\eta > 0$, regularity supplies a compact $K \subset E$ and a relatively open $U \supset E$ with $\nu(U \setminus K) < \eta$. The continuous separation of two disjoint closed sets supplies $f : Q \to [0,1]$ equal to 1 on K and 0 on $Q \setminus U$. It follows that $\int_Q |f - \mathbf{1}_E|\, d\nu < \eta$. Express s_N as a finite sum $\sum_\ell b_\ell \mathbf{1}_{E_\ell}$ over its measurable level sets. Applying the preceding approximation to these sets and summing with the vector coefficients gives continuous fields $h : Q \to \mathcal{E}$ with arbitrarily small $\int_Q \|h - s_N\|_2\, d\nu$. For example, the triangle inequality bounds this integral by $\sum_\ell \|b_\ell\|_2 \int_Q |f_\ell - \mathbf{1}_{E_\ell}|\, d\nu$, and each scalar error can be chosen separately. This also proves the usual density of continuous fields in weighted L^1 after simple-function approximation.

Projection preserves the approximation. Let P be nearest-point projection onto $\mathcal{O}_{L,H}$ in $\mathcal{E}$. To obtain the projection, choose points whose distances from x converge to the infimum. Convexity puts their midpoints in the same set, and the parallelogram identity bounds the squared distance between two chosen points by twice the sum of their squared distances from x minus four times the squared infimum. This tends to zero, so the minimizing sequence is Cauchy. Completeness and closedness give a minimizer p. The same midpoint identity proves uniqueness: two distinct minimizers would have a midpoint with strictly smaller squared distance. Comparing squared distances along the segment from p to any $z \in \mathcal{O}_{L,H}$ gives

$$\langle x - p, z - p \rangle \le 0.$$

For $p = P(x)$ and $q = P(y)$, apply this inequality first with $z = q$ and then with the roles reversed and $z = p$. Adding yields

$$\|p - q\|_2^2 \le \langle p - q, x - y \rangle \le \|p - q\|_2 \, \|x - y\|_2 .$$

Thus P is 1-Lipschitz and continuous, including when $p = q$, and it fixes every point of the dual set.

Given $\varepsilon > 0$, choose the continuous approximation h with $\int_Q \|h - s_N\|_2\, d\nu \le \varepsilon$ and put $u_{N,\varepsilon} = P \circ h$. This is a continuous admissible field. Since $P(s_N) = s_N$ pointwise, nonexpansiveness and $\|v\|_2 \le w$ give

$$\begin{aligned} \int_Q |(s_N - u_{N,\varepsilon}) \cdot v|\, d\rho_\mu &\le \int_Q \|s_N - u_{N,\varepsilon}\|_2\, d\nu \\ &\le \int_Q \|s_N - h\|_2\, d\nu \le \varepsilon. \end{aligned}$$

Every integral here is well-defined: the selector is bounded, continuous fields are bounded on compact Q, and all coordinate densities are integrable. If S denotes the continuous dual

supremum, this estimate implies $\int_Q m_N \, d\rho_\mu \le S + \varepsilon$ for each N. Let $\varepsilon \downarrow 0$ and then use monotone convergence to obtain

$$\mathcal{J}_{L,H}(\mu) = \lim_{N\to\infty} \int_Q m_N \, d\rho_\mu \le S.$$

If the action is infinite, the same increasing selector integrals force $S = +\infty$. This proves the reverse inequality.

For positivity, fix a point at which the symmetric density g has finite local cost. For each $v \in \mathbb{R}^n$ and $s \ge 0$, the triple $(-svv^T, 0, 0)$ is admissible and has pairing $-sv^T gv$. Thus $v^T gv < 0$ implies $j_{L,H}(g, z, r) = +\infty$, a contradiction. All quadratic forms are therefore nonnegative at every such point. Finite action implies finite local cost almost everywhere, hence $g \succeq 0$ almost everywhere. Consequently, for every measurable S and every real v, $v^T G(S) v = \int_S v^T gv \, d\rho_\mu \ge 0$. Finally each continuous dual pairing is weak-star continuous in all coordinates. A supremum of these continuous functions is lower semicontinuous, proving the last assertion. □

Lemma 3.2 (Trace and Frobenius variation estimates)**.** *Let $\mu \in$ CE have finite action $E = \mathcal{J}_{L,H}(\mu)$, set $T = \int_Q \operatorname{tr}(dG)$ and $m_0 = \operatorname{tr} G_0(\Omega)$. Then*

$$T = m_0 + \int_Q (1-t) \operatorname{tr}(dR), \qquad 0 \le T \le 2m_0 + 2\|H\|_F^2 E. \tag{8}$$

For any compatible rectangular A,

$$\|qA\|_{\mathrm{TV},F}^2 \le 2ET \left\|L^T A\right\|_F^2, \qquad \|RA\|_{\mathrm{TV},F}^2 \le 2ET \left\|H^T A\right\|_F^2. \tag{9}$$

All variations in (9) *are finite.*

Proof. Test (1) with $(1-t)I_n$. Since $D^* I_n = 0$, its left side is $-T + \int_Q (1-t)\operatorname{tr}(dR)$ and its right side is $-m_0$, giving the trace identity. Positivity from Lemma 3.1 gives $T \ge 0$. For real s, use the admissible reaction test $C = s(1-t)I_n$, $B = 0$, $A = -CHH^T C^T/2$. Pairing and integrating, then using $(1-t)^2 \le 1$ and $HH^T \preceq \|H\|_F^2 I_n$, gives

$$s(T - m_0) - \tfrac{1}{2} s^2 \|H\|_F^2 T \le E.$$

If $H \neq 0$, substitution of $s = \|H\|_F^{-2}$ gives $(T/2 - m_0)/\|H\|_F^2 \le E$, which is (8). If $H = 0$, the inequality for both signs and arbitrarily large values of s forces $T = m_0$.

For the variation bound, work where the local cost is finite. For a coordinate vector e_i and $v \in \mathbb{R}^k$, use the triple with flux $B = se_i v^T$, reaction $C = 0$ and mass field $-\frac{s^2}{2}\left\|L^T v\right\|^2 e_i e_i^T$. Its pairing is $s(zv)_i - s^2 g_{ii}\left\|L^T v\right\|^2/2$. For real x and $a > 0$,

$$sx - \tfrac{a}{2}s^2 = \frac{x^2}{2a} - \frac{a}{2}\left(s - \frac{x}{a}\right)^2,$$
$$\sup_{s\in\mathbb{R}} \left(sx - \tfrac{a}{2}s^2\right) = \frac{x^2}{2a}.$$

If $a = 0$, finiteness of this supremum forces $x = 0$. Apply these two cases with $a = g_{ii}\left\|L^T v\right\|^2 \ge 0$ and $x = (zv)_i$. In both cases,

$$|(zv)_i|^2 \le 2 j_{L,H}(g, z, r)\, g_{ii} \left\|L^T v\right\|^2.$$

Applying this inequality to each column a_b of the fixed matrix A and summing over i, b gives

$$\begin{aligned}\|zA\|_F^2 &\le 2j_{L,H}(g,z,r)\Big(\sum_i g_{ii}\Big)\Big(\sum_b \left\|L^T a_b\right\|^2\Big)\\ &= 2j_{L,H}(g,z,r)\operatorname{tr}(g)\left\|L^T A\right\|_F^2.\end{aligned}$$

The same rank-one test in the C channel gives $\|rA\|_F^2 \le 2j_{L,H}(g,z,r)\operatorname{tr}(g)\left\|H^T A\right\|_F^2$ for every compatible A. In particular it controls the full reaction density. Integrating square roots and applying Cauchy–Schwarz to the two nonnegative functions $\sqrt{j_{L,H}}$ and $\sqrt{\operatorname{tr} g}$ yields

$$\begin{aligned}\int_Q \|zA\|_F \, d\rho_\mu &\le \sqrt{2}\left\|L^T A\right\|_F \left(\int_Q j_{L,H}\, d\rho_\mu\right)^{1/2}\left(\int_Q \operatorname{tr} g\, d\rho_\mu\right)^{1/2}\\ &= \sqrt{2ET}\left\|L^T A\right\|_F,\end{aligned}$$

and the corresponding reaction inequality. The product densities are integrable; Radon–Nikodym linearity identifies them with the densities of qA and RA. Formula (6) and squaring give (9), including $E = 0$, $T = 0$, and null-weight directions. □

For a positive-definite fixed reaction weight, existence of a minimizer is established in [7, Theorem 4.2]. The following argument treats bounded varying weights and singular fixed weights at finite value.

Lemma 3.3 (Bounded-weight compactness)**.** *If (L_j, H_j) is bounded and admissible μ_j have uniformly bounded actions, a common subsequence converges weak-star in all signed coordinates to an admissible path. For fixed weights, every finite minimum is attained with PSD measure-valued mass.*

Proof. The uniform bounds on the weights and actions, together with (8), bound T_j. Taking coordinate vectors for A in (9) bounds the Frobenius variation of every column of q_j and R_j, hence the variation of every scalar coordinate. Positivity also controls all mass coordinates: the 2×2 principal minors of $g_j \succeq 0$ give

$$|(g_j)_{ab}| \le \sqrt{(g_j)_{aa}(g_j)_{bb}} \le \tfrac{1}{2}\big((g_j)_{aa} + (g_j)_{bb}\big) \le \operatorname{tr} g_j.$$

Integrating shows $|G_j^{ab}|(Q) \le T_j$. Thus every coordinate has a uniform total-variation bound.

We spell out why these bounds give a common subsequence. If Q is empty, every coordinate measure is zero; otherwise choose a point of Q. Decompose each signed coordinate η_j into its positive and negative Jordan parts. Both are finite positive measures with mass bounded by the common variation bound C. A positive measure γ of mass $m \le C$ can be written as $m\pi$, where π is a probability measure; if $m = 0$, any fixed probability measure gives the same zero measure. The space of probability measures on compact metric Q is compact and metrizable for weak convergence. Thus $[0, C]$ times that space is sequentially compact. There are only finitely many coordinates and two Jordan parts for each, so compactness of their finite product supplies a single subsequence along which all masses and all normalized positive measures converge. Multiplication by the masses gives weak limits of all the Jordan parts.

Subtract each limiting negative part from its limiting positive part. The resulting finite signed Radon measure η satisfies $\int f\, d\eta_j \to \int f\, d\eta$ for every $f \in C(Q)$, because this pairing is the difference of the two positive-measure pairings. The limiting positive and negative parts need not remain mutually singular; their difference still has exactly this pairing. Applying the construction simultaneously to all coordinates proves the asserted weak-star convergence.

The relations $G_j^{ab} = G_j^{ba}$ pass to the limit, so the limiting mass is symmetric. For each fixed closed-domain C^1 test in (1), all coordinates of $\partial_t\Phi$, $D^*\Phi$ and Φ are continuous on Q. Every integral in that equation therefore converges, while its prescribed endpoint terms are unchanged. The limiting path is admissible.

For fixed weights and finite infimum $V(L,H)$, choose admissible paths with actions at most $V(L,H) + 1/(j+1)$. This is a bounded finite-action minimizing sequence, so the preceding argument gives an admissible subsequential limit μ. Lemma 3.1 gives $\mathcal{J}_{L,H}(\mu) \le V(L,H)$, and the definition of the infimum gives the opposite inequality. Thus the minimum is attained, and its finite action implies PSD measure-valued mass. □

Lemma 3.4 (Joint lower bound)**.** *If $L_j \to L_*$, $H_j \to H_*$ and $\mu_j \overset{*}{\rightharpoonup} \mu$, then*

$$\mathcal{J}_{L_*,H_*}(\mu) \le \liminf_j \mathcal{J}_{L_j,H_j}(\mu_j). \tag{10}$$

No relative comparison is needed here.

Proof. If the right side of (10) is $+\infty$, the inequality is automatic. Otherwise pass to a subsequence whose finite actions converge to that finite lower limit, denoted by ℓ, and relabel it. Fix a continuous target dual field (A,B,C). We compensate for the change in its quadratic constraint by subtracting a scalar multiple of I_n from its mass field. Let $M_B = \sup_Q \|B\|_F^2$, $M_C = \sup_Q \|C\|_F^2$, write $|S|_1 = \sum_{a,b} |S_{ab}|$, and set

$$\delta_j = \tfrac{1}{2}(M_B |L_j L_j^T - L_* L_*^T|_1 + M_C |H_j H_j^T - H_* H_*^T|_1).$$

For any symmetric S, compatible matrix F and vector x, putting $y = F^T x$ gives

$$|x^T F S F^T x| \le \sum_{a,b} |S_{ab}|\,|y_a|\,|y_b| \le |S|_1 \,\|F\|_F^2\,|x|^2.$$

Consequently $FSF^T \preceq |S|_1 \,\|F\|_F^2\, I_n$. Apply this to the two Gram differences. The change in the quadratic constraint is bounded above by $\delta_j I_n$, uniformly on Q, so $(A - \delta_j I_n, B, C)$ is admissible at (L_j, H_j). Matrix convergence implies $\delta_j \to 0$.

The corrected field has pairing

$$\int_Q A \cdot dG_j + \int_Q B \cdot dq_j + \int_Q C \cdot dR_j - \delta_j \int_Q \operatorname{tr}(dG_j).$$

Weak-star convergence makes the first three terms converge to the target pairing. It also makes the trace pairing converge, since it is a finite sum of coordinate pairings against the constant function 1; thus the last term tends to zero. This step uses no positivity assumption on the limiting mass. The defining support inequality bounds the corrected pairing by $\mathcal{J}_{L_j,H_j}(\mu_j)$, so its limit is at most ℓ. Every continuous target dual pairing is therefore at most ℓ. Taking their supremum in (7) proves $\mathcal{J}_{L_*,H_*}(\mu) \le \ell$, including the assertion that a finite lower limit cannot coexist with infinite target action. □

Lemma 3.5 (Relative Gram comparison)**.** *If $\alpha > 0$ and*

$$LL^T \succeq \alpha L_* L_*^T, \qquad HH^T \succeq \alpha H_* H_*^T, \tag{11}$$

then for every path μ,

$$\mathcal{J}_{L,H}(\mu) \le \alpha^{-1} \mathcal{J}_{L_*,H_*}(\mu). \tag{12}$$

Proof. For $(A,B,C) \in \mathcal{O}_{L,H}$, the target slack of $(\alpha A, \alpha B, \alpha C)$ is

$$\alpha(-A - \tfrac{1}{2}BLL^TB^T - \tfrac{1}{2}CHH^TC^T) + \tfrac{\alpha}{2}B(LL^T - \alpha L_*L_*^T)B^T + \tfrac{\alpha}{2}C(HH^T - \alpha H_*H_*^T)C^T.$$

Each term is PSD, so multiplication by α sends the current dual set into the target dual set and multiplies its pairing by α. Taking support suprema and integrating proves (12), including infinite actions. □

4 Convergent and singular weights

Theorem 4.1 (Simultaneous relative-Gram stability)**.** *Suppose $L_j \to L_*$, $H_j \to H_*$ and $\alpha_j \to 1$, where eventually $\alpha_j > 0$ and*

$$L_j L_j^T \succeq \alpha_j L_* L_*^T, \qquad H_j H_j^T \succeq \alpha_j H_* H_*^T. \tag{13}$$

Then $V(L_j, H_j) \to V(L_*, H_*)$ in $[0, +\infty]$. If $v_* = V(L_*, H_*) < \infty$, the target minimum is attained with PSD mass. For any admissible sequence satisfying

$$\mathcal{J}_{L_j,H_j}(\mu_j) \le V(L_j, H_j) + \eta_j, \qquad \eta_j \ge 0, \quad \eta_j \to 0,$$

the actions tend to v_*. At finite v_* it has a common weak-star subsequence converging to a target minimizer, every weak-star subsequence limit is a PSD target minimizer, and along the whole sequence

$$L_*^T A = 0 \Rightarrow \|q_j A\|_{\mathrm{TV},F} \to 0, \qquad H_*^T A = 0 \Rightarrow \|R_j A\|_{\mathrm{TV},F} \to 0. \tag{14}$$

The action convergence also holds when $v_* = +\infty$.

Proof. If $v_* < \infty$, Lemma 3.3 gives a target minimizer μ_*. For this fixed admissible path, the Gram comparison gives, eventually,

$$V(L_j, H_j) \le \mathcal{J}_{L_j,H_j}(\mu_*) \le \alpha_j^{-1} v_*.$$

Thus $\limsup_j V(L_j, H_j) \le v_*$.

For the lower bound, suppose that $\ell = \liminf_j V(L_j, H_j) < \infty$. Pass to a subsequence along which the minima converge to ℓ through finite values. For each index choose an admissible path with action between that minimum and that minimum plus $1/(j+1)$; such a finite-action path exists by the definition of the infimum. Their actions converge to ℓ and are eventually bounded. Lemma 3.3 gives a further weak-star subsequence with admissible limit μ, and Lemma 3.4 gives

$$v_* \le \mathcal{J}_{L_*,H_*}(\mu) \le \ell.$$

This proves the lower bound when v_* is finite. If $v_* = +\infty$, the same argument rules out a finite lower limit, so the minima tend to $+\infty$.

Near-minimizer actions are squeezed between the minima and their vanishing-gap upper bounds. At finite v_* they are eventually bounded, so Lemma 3.3 provides a weak-star subsequence. Every weak-star cluster is admissible, and Lemma 3.4 bounds its target action by v_*; hence it is a target minimizer with PSD mass. Finally (8) bounds the traces along the whole tail of the sequence. In (9), the energy and trace factors are therefore bounded, while $L_j^T A \to 0$ or $H_j^T A \to 0$ in each forbidden target direction. This proves (14) along the whole sequence. □

Corollary 4.2 (Independent fixed eigenbases)**.** *Let $U \in O(k)$ and $W \in O(n)$ be fixed, and suppose all nonnegative coefficients in*

$$L_j = U \operatorname{diag}(\ell_j) U^T, \qquad H_j = W \operatorname{diag}(h_j) W^T$$

converge to their target coefficients. Then all conclusions of Theorem 4.1 hold, without a separate comparison assumption. The bases may differ, and the coefficients need not be monotone.

Proof. For each nonzero target coefficient a_* let its error be $|a_j^2/a_*^2 - 1|$, and for zero target coefficients let its error be zero. Let δ_j be the sum of these errors over both weights, and set $\alpha_j = \max(1/2, 1 - \delta_j)$. Then $\alpha_j \to 1$, and eventually $a_j^2 \geq \alpha_j a_*^2$ for every coefficient. The zero target entries require no restriction. Orthogonal congruence gives (13). □

Corollary 4.3 (Reaction zero and partial reaction)**.** *Fix L. If $H_j \to 0$, then*

$$V(L, H_j) \longrightarrow b_L := \inf\{\mathcal{J}_{L,I_n}(G, q, 0) : (G, q, 0) \in \mathrm{CE}\}.$$

This includes $b_L = +\infty$. At finite b_L, balanced minimizers exist, near-minimizer actions tend to b_L, every weak-star cluster is balanced and optimal, and $\|R_j\|_{\mathrm{TV},F} \to 0$ along the entire sequence. More generally, if $H_j \to H_*$ and eventually $H_j H_j^T \succeq \alpha_j H_* H_*^T$ with $0 < \alpha_j \leq 1$ and $\alpha_j \to 1$, Theorem 4.1 holds with fixed L. The surviving reaction directions remain in the target problem.

Proof. For $H_* = 0$ choose $\alpha_j = 1$. Finite action at zero reaction weight forces $R = 0$: with $H = 0$, the triples $(0, 0, C)$ are admissible for every matrix C, so a nonzero local reaction density makes the support supremum infinite. For $R = 0$, deleting C preserves feasibility and leaves the support pairing unchanged, while the $C = 0$ slice is independent of H. Thus the action is independent of H on these paths, and the target infimum is exactly b_L. For partial reaction, the fixed transport Gram dominates $\alpha_j L L^T$. Apply Theorem 4.1 in both cases; use $A = I_n$ for full reaction extinction in the zero case. □

Corollary 4.4 (Both zero weights)**.** *If $L_j \to 0$ and $H_j \to 0$, all conclusions of Theorem 4.1 hold with $(L_*, H_*) = (0, 0)$, irrespective of rotating eigenbases. The same statement applies to zero transport with a fixed reaction weight. For a positive-definite fixed reaction weight, this recovers the zero-transport value and minimizer limit of [7, Proposition 4.6]. Theorem 4.1 also covers singular reaction weights, vanishing-gap near-minimizers, and extinction in each null direction of the limiting weights.*

Proof. The zero target Grams satisfy the comparison with factor one automatically. For fixed reaction, its Gram comparison is equality. □

5 Large transport weights

The large-transport regimes below keep the reaction weight fixed.

Define the reaction-channel action by

$$F_H(\mu) = \int_Q j_{L,H}(g, 0, r)\, d\rho_\mu.$$

At zero flux density, deleting B from an admissible dual triple preserves feasibility and leaves its pairing unchanged. Conversely, every admissible triple with $B = 0$ belongs to the full dual set. Thus the support supremum at $(g, 0, r)$ equals its supremum over the $B = 0$ slice, which is independent of L. This also proves $F_H(\mu) \leq \mathcal{J}_{L,H}(\mu)$, because the same slice is available when the flux density is nonzero. Lemma 2.1 justifies computing F_H using ρ_μ. It retains the mass and reaction range constraints, while μ still satisfies the continuity equation with its full flux q. For a fixed baseline transport weight put

$$W(L, H) = \inf\{F_H(\mu) : \mu \in \mathrm{CE},\ \mathcal{J}_{L,H}(\mu) < \infty\}. \tag{15}$$

The finite-baseline requirement retains the transport range and integrability constraints of the original problem.

Lemma 5.1 (Ray recovery and finite-domain preservation)**.** *For $t \geq 1$ and every finite-baseline path,*

$$F_H(\mu) \leq \mathcal{J}_{tL,H}(\mu) \leq F_H(\mu) + t^{-1}\mathcal{J}_{L,H}(\mu), \qquad \mathcal{J}_{L,H}(\mu) \leq t^2 \mathcal{J}_{tL,H}(\mu). \tag{16}$$

In particular the baseline and current finite-action classes coincide.

Proof. Write $v = (g, z, r)$ and $v_s = (g, sz, r)$. The dual substitution $B \mapsto tB$ gives $j_{tL,H}(v) = j_{L,H}(v_{1/t})$. If (A, B, C) is admissible, then $(A, 0, C)$ is also admissible and has the same pairing with $(g, 0, r)$. Taking support suprema therefore gives $j_{L,H}(g, 0, r) \leq j_{L,H}(v_{1/t})$. Convexity gives

$$j_{L,H}(v_{1/t}) \leq (1 - t^{-1}) j_{L,H}(g, 0, r) + t^{-1} j_{L,H}(v) \leq j_{L,H}(g, 0, r) + t^{-1} j_{L,H}(v).$$

Integrating against ρ_μ proves the first two inequalities in (16).

For the last inequality, apply Lemma 3.5 with the baseline weights as the current weights, the weights (tL, H) as the target, and $\alpha = t^{-2}$. The transport comparison is equality and the reaction comparison follows from $t \geq 1$. This gives $\mathcal{J}_{L,H}(\mu) \leq t^2 \mathcal{J}_{tL,H}(\mu)$ for every path μ. Hence finite current action implies finite baseline action. Conversely, the first two inequalities, together with $F_H \leq \mathcal{J}_{L,H}$, show that finite baseline action implies finite current action. □

Theorem 5.2 (Fixed ray, including singular rays)**.** *For any fixed L, H and $0 < t_j \to +\infty$,*

$$V(t_j L, H) \longrightarrow W(L, H)$$

in the extended nonnegative reals.

Proof. Eventually the finite-action class is the baseline class. Every current action on this class dominates F_H, giving $V(t_j L, H) \geq W(L, H)$. If $W(L, H) < \infty$, for each $\varepsilon > 0$ choose a finite-baseline competitor μ_ε with $F_H(\mu_\varepsilon) \leq W(L, H) + \varepsilon$, as permitted by the definition of infimum. Keeping this path fixed, (16) gives

$$\limsup_j V(t_j L, H) \leq F_H(\mu_\varepsilon) + \lim_j t_j^{-1} \mathcal{J}_{L,H}(\mu_\varepsilon) \leq W(L, H) + \varepsilon.$$

Its baseline action is a finite constant; taking $\varepsilon \downarrow 0$ therefore gives the upper bound. If the target is infinite, the lower bound is already conclusive. □

Theorem 5.3 (Uniform divergence in every direction)**.** *Suppose $t_j \to +\infty$ and eventually $L_j L_j^T \succeq t_j^2 I_k$. Then*

$$V(L_j, H) \longrightarrow W(I_k, H).$$

For admissible vanishing-gap near-minimizers, their actions converge to this extended target. If the target is finite, their reaction-channel actions also converge to it.

Proof. For each j, the Gram bound $L_j L_j^T \preceq (1 + \|L_j\|_F^2) I_k$ permits Lemma 3.5 with factor $(1 + \|L_j\|_F^2)^{-1}$, giving

$$\mathcal{J}_{I_k,H}(\mu) \leq (1 + \|L_j\|_F^2) \mathcal{J}_{L_j,H}(\mu).$$

Thus finite current action implies finite identity-baseline action. The comparison constant may depend on j, since this step concerns finiteness at each index. Conversely, eventually $t_j \geq 1$ and $\mathcal{J}_{L_j,H}(\mu) \leq \mathcal{J}_{t_j I_k,H}(\mu)$ by the assumed Gram domination. Lemma 5.1 makes the latter finite on the identity-baseline class. The two finite-action classes therefore coincide eventually.

For a fixed baseline path,

$$F_H(\mu) \le \mathcal{J}_{L_j,H}(\mu) \le \mathcal{J}_{t_j I_k,H}(\mu) \to F_H(\mu).$$

The infimum argument of Theorem 5.2 now gives the value limit. Squeezing between the minimum values and their vanishing-gap upper bounds proves near-minimizer action convergence. At finite target these actions are eventually finite, so the same paths have finite baseline action. Consequently $W(I_k, H) \le F_H(\mu_j) \le \mathcal{J}_{L_j,H}(\mu_j)$, which proves the reaction-channel action convergence. □

6 Entrywise divergence and the Bures value

In this section $k = nd$, and $q = (Q_1, \dots, Q_d)$ is the horizontal concatenation of $n \times n$ matrix-valued measure blocks. We use $D^*\Phi = (\partial_1\Phi, \dots, \partial_d\Phi)$. The continuity equation remains the weak formulation (1). For smooth admissible densities with $F = (F_1, \dots, F_d)$, the block divergence and continuity equation are

$$(\operatorname{Div} F)_{ij} = \sum_{a=1}^{d} \partial_a (F_a)_{ij}, \qquad \partial_t M + \sum_{a=1}^{d} \partial_a \operatorname{sym} F_a = \operatorname{sym} Z.$$

The second identity is the density form of (1) in the interior. For an $n \times d$ matrix field Φ, row divergence instead means $(\operatorname{div}\Phi)_i = \sum_{a=1}^{d} \partial_a \Phi_{ia}$; it is the test-side operator used in the Kantorovich–Bures examples. Every spatially constant symmetric test is annihilated by D^*, a property stronger than $D^* I_n = 0$. Fix $H \in \mathbb{S}^n_{++}$.

For PSD matrices define the usual Bures–Wasserstein expression

$$d_B^2(X, Y) = \operatorname{tr} X + \operatorname{tr} Y - 2 \operatorname{tr} \sqrt{X^{1/2} Y X^{1/2}}.$$

This expression and the orthogonal Procrustes characterization are established matrix geometry [1]. Weighted reaction congruence is already present in [7, Remark 3.11] and is related to [5].

Lemma 6.1 (Pure reaction with PSD endpoints)**.** *Let σ be a probability Radon measure on Ω and $A_0, A_1 \succeq 0$. For endpoints $A_0\sigma, A_1\sigma$, the minimum of the original action over admissible paths with $q = 0$ equals*

$$E_H(A_0, A_1) = 2 d_B^2(H^{-1} A_0 H^{-1}, H^{-1} A_1 H^{-1}). \tag{17}$$

The matrices A_i may be singular, and the minimum is attained.

Proof. Write $X_i = H^{-1} A_i H^{-1}$. Choose factors $P_i P_i^T = X_i$ with $P_0 = X_0^{1/2}$ and $\|P_1 - P_0\|_F^2 = d_B^2(X_0, X_1)$. Here is an explicit construction covering singular endpoints. For $\delta > 0$ put

$$S_\delta = (X_0 + \delta I)^{1/2}, \qquad K_\delta = (S_\delta X_1 S_\delta)^{1/2}, \qquad P_\delta = S_\delta^{-1} K_\delta.$$

Then $P_\delta P_\delta^T = S_\delta^{-1} K_\delta^2 S_\delta^{-1} = X_1$ and $\operatorname{tr}(S_\delta P_\delta) = \operatorname{tr} K_\delta$. In particular $\|P_\delta\|_F^2 = \operatorname{tr} X_1$, so finite-dimensional compactness supplies a convergent subsequence as $\delta \downarrow 0$; denote its limit by P_1. The PSD square-root map is continuous: for a convergent sequence of PSD matrices its square roots are bounded, and every subsequential limit is PSD and squares to the limiting matrix, whose PSD square root is unique. Consequently the factor and trace identities pass to the limit, giving

$$P_1 P_1^T = X_1, \qquad \operatorname{tr}(P_0 P_1) = \operatorname{tr}\big((X_0^{1/2} X_1 X_0^{1/2})^{1/2}\big).$$

Expanding $\|P_1 - P_0\|_F^2 = \operatorname{tr} X_1 + \operatorname{tr} X_0 - 2\operatorname{tr}(P_0 P_1)$ proves the required identity and the nonnegativity of d_B^2.

Put $\Delta = P_1 - P_0$, $P_s = P_0 + s\Delta$, and

$$M_s = H P_s P_s^T H, \qquad Z_s = 2H P_s \Delta^T H.$$

Differentiation gives $\dot{M}_s = H(\Delta P_s^T + P_s \Delta^T)H = \operatorname{sym} Z_s$. The path $(M_s\, ds \otimes \sigma, 0, Z_s\, ds \otimes \sigma)$ has endpoints $A_i\sigma$ and satisfies the original weak equation. Symmetric tests pair with Z_s through $\operatorname{sym} Z_s$. For interior x, the product rule and time fundamental theorem applied to $\Phi(s,x) \cdot M_s$ give the endpoint identity. For $x \in \partial\Omega$, choose interior points $x_m \to x$, possible because Ω is the closure of its interior. Uniform continuity of Φ and $\partial_s \Phi$ on compact Q gives uniform convergence in s along these points. Since M_s and $\dot{M}_s$ are bounded, the interior identity passes to the limit. Integrating this identity in the finite measure σ proves (1) also for boundary-supported σ. For an admissible dual triple (A, B, C), the nonnegative transport Gram term may be dropped after pairing its constraint with M_s. With $Y = (CH)^T H P_s$, cyclicity of trace gives $\operatorname{tr}(CH^2 C^T M_s) = \|Y\|_F^2$ and $C \cdot Z_s = 2\langle Y, \Delta \rangle_F$. Thus

$$A \cdot M_s + C \cdot Z_s \le -\tfrac{1}{2}\|Y\|_F^2 + 2\langle Y, \Delta \rangle_F = 2\|\Delta\|_F^2 - \tfrac{1}{2}\|Y - 2\Delta\|_F^2 \le 2\|\Delta\|_F^2 .$$

Integrating against $ds \otimes \sigma$ and taking the continuous-dual supremum in (7) proves that this path has pure-reaction action at most $2d_B^2(X_0, X_1)$. The estimate applies even where M_s is singular.

For the matching lower bound, regularize both endpoints and set

$$S_\delta = (X_0 + \delta I)^{1/2}, \qquad T_\delta = S_\delta^{-1}[S_\delta(X_1 + \delta I)S_\delta]^{1/2} S_\delta^{-1}.$$

The symmetric matrix T_δ is positive definite. Thus $N_s = (1-s)I + sT_\delta$ is invertible for $s \in [0,1]$, and

$$\Psi_s = 2(T_\delta - I)N_s^{-1}, \qquad \Phi_s = H^{-1}\Psi_s H^{-1}$$

are time-dependent symmetric tests. Indeed, N_s is a polynomial in T_δ, so N_s^{-1} commutes with $T_\delta - I$. Differentiating $N_s N_s^{-1} = I$ yields

$$\dot{\Psi}_s = -2(T_\delta - I)N_s^{-1}(T_\delta - I)N_s^{-1} = -\tfrac{1}{2}\Psi_s^2.$$

Since $H^{-1}H^2H^{-1} = I$, congruence by H^{-1} gives $\dot{\Phi}_s + \Phi_s H^2 \Phi_s/2 = 0$. Hence $(\dot{\Phi}_s, 0, \Phi_s)$ satisfies the dual constraint. The original weak equation and the defining support inequality bound every pure-reaction competitor below by $\Phi_1 \cdot A_1 - \Phi_0 \cdot A_0$; the spatial integrals reduce to these pairings because $\sigma(\Omega) = 1$. Write $\widehat{K}_\delta = [S_\delta(X_1 + \delta I)S_\delta]^{1/2}$ and $c_\delta = \operatorname{tr}\widehat{K}_\delta$. Direct multiplication gives

$$T_\delta S_\delta^2 T_\delta = S_\delta^{-1}\widehat{K}_\delta^2 S_\delta^{-1} = X_1 + \delta I, \qquad \operatorname{tr}(T_\delta S_\delta^2) = \operatorname{tr}\widehat{K}_\delta = c_\delta.$$

Also $T_\delta^{-1}(X_1 + \delta I) = S_\delta^2 T_\delta$, whose trace is c_δ. Subtracting the regularization terms therefore gives

$$\operatorname{tr}(T_\delta X_0) = c_\delta - \delta \operatorname{tr} T_\delta, \qquad \operatorname{tr}(T_\delta^{-1} X_1) = c_\delta - \delta \operatorname{tr} T_\delta^{-1}.$$

Since $\Psi_0 = 2(T_\delta - I)$ and $\Psi_1 = 2(I - T_\delta^{-1})$, the endpoint pairing is the sum of

$$2\{\operatorname{tr} X_0 + \operatorname{tr} X_1 - 2\operatorname{tr}([S_\delta(X_1 + \delta I)S_\delta]^{1/2})\}$$

and $2\delta \operatorname{tr}(T_\delta + T_\delta^{-1}) \ge 0$. Drop this nonnegative second term before passing to the limit. The first term converges to $2d_B^2(X_0, X_1)$ by continuity of the PSD square root, so every competitor has at least that action. This proves the lower bound, including singular endpoints. The same space-constant certificate gives the lower bound for any original transport problem whose operator annihilates all spatially constant symmetric fields. □

A noncommuting inflation example. We compare the inflation expression in [7, Proposition 4.4, equation (4.12)] with the variational value for the following noncommuting pair. Take $x_0 \in \operatorname{int}\Omega$, $e = (1,0)^T$, $v = (1,1)^T/\sqrt{2}$, $P_0 = ee^T$, $P_1 = vv^T$, and $H = I_2$. For $\mathsf{G} = I_2\delta_{x_0}$, the inflated endpoints $\mathsf{G}^{P_i} := P_i\mathsf{G}P_i$ equal $P_i\delta_{x_0}$. For these endpoints, that expression equals $2\operatorname{tr}((P_1 - P_0)^2) = 2$. Put $w = v - e$ and $u_s = e + sw$. Since $|u_s|^2 = 1 - (2-\sqrt{2})s(1-s) > 0$, the zero-flux path with mass $u_s u_s^T\, ds \otimes \delta_{x_0}$ and reaction $2u_s w^T\, ds \otimes \delta_{x_0}$ is admissible: $\partial_s(u_s u_s^T) = \operatorname{sym}(2u_s w^T)$. Its local support cost is

$$j_{L,I_2}(u_s u_s^T, 0, 2u_s w^T) = \sup_C \left\{ 2w \cdot C^T u_s - \tfrac{1}{2}|C^T u_s|^2 \right\} = 2|w|^2 = 4 - 2\sqrt{2} < 2.$$

Indeed, the maximizing choice of the mass dual variable saturates its constraint, the transport dual variable can be set to zero, and $C^T u_s$ ranges over $\mathbb{R}^2$. For entrywise divergence, Lemma 6.1 and its space-constant lower certificate give the exact minimum $2d_B^2(P_0, P_1) = 4 - 2\sqrt{2} < 2$.

Theorem 6.2 (Radon hub recovery)**.** *Suppose Ω is convex, choose $x_* \in \operatorname{int}\Omega$, and put $A_i = G_i(\Omega)$,*

$$M_i = \int_\Omega |x - x_*|^2\, d\operatorname{tr} G_i(x), \qquad B = (\sqrt{M_0} + \sqrt{M_1})/\sqrt{2}.$$

For arbitrary finite PSD Radon endpoints and every $t > 0$,

$$E_H(A_0, A_1) \le V(tI_{nd}, H) \le \left(\sqrt{E_H(A_0, A_1)} + B/t\right)^2. \tag{18}$$

Consequently $W(I_{nd}, H) = E_H(A_0, A_1)$. For SPD L with $L^2 \succeq m^2 I_{nd}$, $m > 0$, the same bound holds with t replaced by m.

Proof. Throughout this proof abbreviate $E_H := E_H(A_0, A_1)$. The space-constant certificates in Lemma 6.1 are admissible for entrywise divergence. They bound every original competitor below by $E_H(A_0, A_1)$, since their endpoint pairings depend only on the total matrices A_i. This proves the lower bound.

For the upper bound, we contract the endpoints to a common hub and connect their total matrices there by pure reaction. For a single endpoint G, take for λ the sum of its coordinate total variations and write $G = N\lambda$. Since $G^{ab} = G^{ba}$, uniqueness of Radon–Nikodym derivatives gives $N = N^T$ almost everywhere. For each vector v in a countable dense subset of $\mathbb{R}^n$, positivity of the scalar measure $v^T G v$ implies $v^T N v \ge 0$ almost everywhere. Removing the union of these countably many null sets and using continuity in v gives $N \succeq 0$ almost everywhere. Define $F_s(x) = (1-s)x + sx_*$, $v(x) = x_* - x$, and $\mathcal{F}(s, x) = (s, F_s(x))$. The contracted measures are exactly

$$\widehat{G} = \mathcal{F}_\#(N\, ds\, d\lambda), \qquad \widehat{Q}_\ell = \mathcal{F}_\#(v_\ell N\, ds\, d\lambda), \qquad \widehat{R} = 0.$$

Their coordinates are finite signed Radon measures: v is bounded on compact Ω, every coordinate of N is integrable, and continuous pushforward preserves finiteness and the Radon property on compact spaces. For a closed-domain test Φ, the time derivative of $\Phi(s, F_s(x))$ is $\partial_s\Phi + \sum_\ell v_\ell \partial_\ell \Phi$ at $(s, F_s(x))$. To justify this also when $x \in \partial\Omega$, choose $B(x_*, r) \subset \operatorname{int}\Omega$. Convexity gives $B(F_s(x), sr) \subset \Omega$ for $s > 0$, so the ordinary chain rule applies there; the continuous extensions of the derivatives make it valid in the interval fundamental theorem down to $s = 0$. Pairing with N and integrating therefore gives

$$\int_Q \partial_s\Phi \cdot d\widehat{G} + \sum_\ell \partial_\ell\Phi \cdot d\widehat{Q}_\ell = \int_\Omega [\Phi(1, x_*) - \Phi(0, x)] \cdot N(x)\, d\lambda(x).$$

Fubini is applicable because the derivatives and v are bounded and $\operatorname{tr} N$ is integrable. This is precisely (1) from G to $G(\Omega)\delta_{x_*}$, including boundary atoms.

To bound the pre-pushforward cost, factor $N = SS^T$. For an admissible dual triple $(A, (B_\ell)_\ell, C)$, dropping the reaction Gram term and completing the square give

$$A \cdot N + \sum_\ell B_\ell \cdot (v_\ell N) \le \sum_\ell \left(-\frac{t^2}{2} \left\| B_\ell^T S \right\|_F^2 + v_\ell \left\langle S, B_\ell^T S \right\rangle_F \right) \le \frac{|v|^2 \operatorname{tr} N}{2t^2}.$$

Here $\left\langle S, B_\ell^T S \right\rangle_F = B_\ell \cdot N$ and $\|S\|_F^2 = \operatorname{tr} N$, so the second inequality is the scalar-vector square completion with maximizing value $v_\ell^2 \|S\|_F^2 / (2t^2)$. Lemma 2.1 permits integration against $ds \otimes \lambda$. For completeness, if (A, B, C) is any continuous admissible field on the image space, its composition with $\mathcal{F}$ is a continuous admissible field on the source: the dual constraint depends only on its values and on the fixed weights. The signed pushforward identity gives, for example,

$$\int_Q A \cdot d\widehat{G} + \sum_\ell B_\ell \cdot d\widehat{Q}_\ell = \int_Q \left[(A \circ \mathcal{F}) \cdot N + \sum_\ell (B_\ell \circ \mathcal{F}) \cdot v_\ell N \right] ds \, d\lambda.$$

The preceding pointwise bound and (7) show that the contraction cost is at most $M_G/(2t^2)$, where $M_G = \int |x - x_*|^2 d \operatorname{tr} G$. This argument also covers collisions of distinct source points at the hub. For expansion, run the curve backwards and reverse the flux. The cost is unchanged, since the dual constraint is invariant under $B \mapsto -B$.

We use the time-rescaling rule of [7, Lemma 3.15] to concatenate these paths, with its measure-level form written explicitly below. For a unit-time piece $\mu = (G, q, R)$ and an interval $[a, a + r]$ of length $r > 0$, let $\Theta(s, x) = (a + rs, x)$ and set

$$G' = r\Theta_\# G, \qquad q' = \Theta_\# q, \qquad R' = \Theta_\# R.$$

For a closed-domain test Φ, $\Psi = \Phi \circ \Theta$ satisfies $\partial_s \Psi = r(\partial_t \Phi) \circ \Theta$ and $D^* \Psi = (D^* \Phi) \circ \Theta$. Substitution in the original equation proves the compressed weak equation with the same spatial endpoints at times $a, a + r$. If (g, z, ζ) are densities of μ with respect to ρ, the densities of the compressed path with respect to $\Theta_\# \rho$ are $(rg, z, \zeta) \circ \Theta^{-1}$. Reference-measure independence and (5) give the exact identity

$$\mathcal{J}_{tI,H}(G', q', R') = \int j_{tI,H}(rg, z, \zeta) \, d\rho = r^{-1} \mathcal{J}_{tI,H}(G, q, R).$$

Apply this operation to the first contraction, the pure-reaction path at the hub with $\sigma = \delta_{x_*}$, and the second expansion, using consecutive intervals of lengths $r_0, r_1, r_2 > 0$ summing to one. Each piece has a Lebesgue factor in time and hence assigns zero to its endpoint slices. Their coordinate sums therefore have no duplicated junction mass. Summing the three weak equations cancels their two internal endpoint pairings and gives (1). Off the junction slices the measures have disjoint time supports, so reference-measure independence makes the action of the sum the sum of the piece actions. Consequently the action is at most

$$\frac{M_0}{2t^2 r_0} + \frac{E_H}{r_1} + \frac{M_1}{2t^2 r_2}.$$

For the three nonnegative costs $(c_0, c_1, c_2) = (M_0/(2t^2), E_H, M_1/(2t^2))$, Cauchy–Schwarz gives $(\sum_i \sqrt{c_i})^2 \le (\sum_i c_i / r_i)(\sum_i r_i)$. Thus the infimum over positive durations summing to one is $(\sqrt{E_H} + B/t)^2$. If all three costs are positive, it is attained by durations proportional to their square roots; if some, but not all, vanish, it is approached by positive durations whose zero-cost components tend to zero; if all three vanish, every such choice gives zero. Taking infima gives the upper bound. Along $t \to \infty$, the two bounds force $V(tI_{nd}, H) \to E_H$. The fixed-ray limit in Theorem 5.2 identifies the same limit as $W(I_{nd}, H)$. For the general SPD bound, $LL^T \succeq m^2 I_{nd}$ and the unchanged reaction Gram imply $\mathcal{J}_{L,H} \le \mathcal{J}_{mI,H}$ by the support-set comparison with $\alpha = 1$. □

7 Rotations and limitations

The following exact examples delineate the scope of the preceding limit theorems. Throughout this section, the planar rotation is

$$U_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Proposition 7.1 (An exactly solvable rotating reaction family)**.** *Use entrywise divergence on a unit-volume domain, uniform endpoints $G_0 = I_2\,dx$, $G_1 = \mathrm{diag}(2,1)\,dx$, and any fixed PSD transport weight. Put $H_{\varepsilon,\theta} = U_\theta\,\mathrm{diag}(1,\varepsilon)U_\theta^T$, where U_θ is the planar rotation, $\varepsilon > 0$. The singular target $H_* = \mathrm{diag}(1,0)$ has value $V_* = 2(\sqrt{2}-1)^2$. Writing $p = \sin^2\theta$ and $Z = 1 + p + 2\sqrt{2}\varepsilon^2 + (2-p)\varepsilon^4$, the current value is*

$$V = 2(3-p) + \frac{2(p/\varepsilon)^2}{(\sqrt{1+p}+1)^2} - \frac{4[2\sqrt{2} + (2-p)\varepsilon^2]}{\sqrt{Z} + \sqrt{1+p}}. \tag{19}$$

As $\varepsilon \to 0$ and $\theta \to 0$, an extended limit $\sin^2\theta/\varepsilon \to \ell$ implies $V \to V_* + \ell^2/2$. In particular $\theta = o(\sqrt{\varepsilon})$ gives $V \to V_*$, $\theta/\sqrt{\varepsilon} \to a$ gives $V \to V_* + a^4/2$, and $|\theta|/\sqrt{\varepsilon} \to \infty$ gives $V \sim \theta^4/(2\varepsilon^2) \to \infty$.

Proof. The factor construction in Lemma 6.1, applied to the uniform endpoints, gives an admissible zero-flux path with action $2d_B^2(X_0, X_1)$, where $A_0 = I_2$, $A_1 = \mathrm{diag}(2,1)$ and $X_i = H^{-1}A_iH^{-1}$. Its space-constant Riccati certificates give the matching lower bound against all competitors. Thus this is the original minimum for any transport weight.

If the eigenvalues of a two-dimensional PSD matrix Y are $\lambda_1, \lambda_2 \geq 0$, then $(\mathrm{tr}\sqrt{Y})^2 = (\sqrt{\lambda_1} + \sqrt{\lambda_2})^2 = \mathrm{tr}\,Y + 2\sqrt{\det Y}$. In the eigenbasis of H, one has

$$U_\theta^T A_1 U_\theta = \begin{pmatrix} 2-p & -\sin\theta\cos\theta \\ -\sin\theta\cos\theta & 1+p \end{pmatrix}, \qquad U_\theta^T X_0 U_\theta = \mathrm{diag}(1, \varepsilon^{-2}).$$

Congruence of the first matrix by $\mathrm{diag}(1,\varepsilon^{-1})$ gives $U_\theta^T X_1 U_\theta$. Taking traces and determinants yields

$$\begin{aligned} \mathrm{tr}\,X_0 + \mathrm{tr}\,X_1 &= 3 - p + (2+p)/\varepsilon^2, \\ \mathrm{tr}(X_0^{1/2} X_1 X_0^{1/2}) &= 2 - p + (1+p)/\varepsilon^4, \\ \det(X_0^{1/2} X_1 X_0^{1/2}) &= 2/\varepsilon^4. \end{aligned}$$

The square-root trace is $\sqrt{Z}/\varepsilon^2$, so substitution gives

$$V = 2(3-p) + \frac{2}{\varepsilon^2}(2 + p - 2\sqrt{Z}).$$

To expose the cancellation, first split $\sqrt{Z}$ at $\sqrt{1+p}$ and use

$$\sqrt{Z} - \sqrt{1+p} = \frac{2\sqrt{2}\varepsilon^2 + (2-p)\varepsilon^4}{\sqrt{Z} + \sqrt{1+p}}, \qquad 2 + p - 2\sqrt{1+p} = \frac{p^2}{(\sqrt{1+p}+1)^2}.$$

These two identities give (19). Its first and last terms together tend to $6 - 4\sqrt{2} = V_*$ and remain bounded, since $p \to 0$ and both square roots in the last denominator tend to one. The middle term tends to $\ell^2/2$ if $p/\varepsilon \to \ell < \infty$ and is asymptotic to $p^2/(2\varepsilon^2)$ if that ratio diverges. In the latter case it dominates the bounded terms. Finally $\sin\theta/\theta \to 1$ converts these conclusions into each of the stated thresholds in θ. For the singular target the uniform

density $M(t) = \mathrm{diag}((1 + (\sqrt{2} - 1)t)^2, 1)$, with $G = M(t)\,dt\,dx$, $R = \dot{M}(t)\,dt\,dx$ and $q = 0$, is admissible by integration by parts in time. Put $a = \sqrt{2} - 1$. Its sole nonzero reaction entry is $2a(1 + at)$ in the supported first column. To compute the cost, maximize first over A by setting $A = -BLL^TB^T/2 - CH_*^2C^T/2$. The zero flux makes $B = 0$ optimal. Writing $m = (1 + at)^2$ and $z = 2a(1 + at)$, the remaining objective is

$$zC_{11} - \tfrac{1}{2}mC_{11}^2 - \tfrac{1}{2}C_{21}^2 \le \frac{z^2}{2m} = 2a^2.$$

Equality holds at $C_{11} = z/m$, $C_{21} = 0$, with the unused second column set to zero. The unit-volume domain therefore gives total action V_*. For the lower bound, $I_2 \succeq H_*^2$ and Lemma 3.5 imply that the identity reaction weight is no more costly than H_*. The latter minimum is thus at least the identity-weight Bures value V_*. This proves equality at the singular target. □

Proposition 7.2 (A rotating transport jump)**.** *In the scalar unit-disk problem with ordinary divergence and $H = 1$, take $G_0 = \delta_{(-1/2,0)}$, $G_1 = \delta_{(1/2,0)}$ and $L_\varepsilon = U_{\sqrt{\varepsilon}}\,\mathrm{diag}(1, \varepsilon)U_{\sqrt{\varepsilon}}^T$. For sufficiently small $\varepsilon > 0$, $V(L_\varepsilon, 1) = 4$, whereas $V(\mathrm{diag}(1, 0), 1) \le 1/2$.*

Proof. Write $x_0 = (-1/2, 0)$ and $x_1 = (1/2, 0)$. A dual cutoff in the expensive direction gives the lower bound; two stationary reaction paths attain it. Let $u = (\sin\sqrt{\varepsilon}, -\cos\sqrt{\varepsilon})$, the expensive unit eigendirection. Define the globally C^1 function

$$h(z) = \begin{cases} 0 & z \le -\pi/2, \\ (1 + \sin z)/2 & |z| < \pi/2, \\ 1 & z \ge \pi/2, \end{cases} \qquad f(x) = h(u \cdot x/\varepsilon).$$

At both breakpoints the interior derivative $\cos z/2$ vanishes, matching the constant pieces; hence h is C^1. Moreover $(h')^2 = h(1 - h)$ on each piece and at the breakpoints. Since $L_\varepsilon u = \varepsilon u$,

$$|L_\varepsilon \nabla f|^2 = |L_\varepsilon u|^2 \varepsilon^{-2}(h'(u \cdot x/\varepsilon))^2 = f(1 - f).$$

The values $u \cdot x_0 = -\sin\sqrt{\varepsilon}/2$ and $u \cdot x_1 = \sin\sqrt{\varepsilon}/2$ lie on opposite constant plateaus once $\sin\sqrt{\varepsilon} \ge \pi\varepsilon$. This holds eventually, because $\sin\sqrt{\varepsilon}/\varepsilon \to \infty$, and gives $f(x_0) = 0$, $f(x_1) = 1$. For $\delta > 0$ set $a(t) = -2/(1 + \delta - t)$, $b(t) = 2/(t + \delta)$, $\Phi = (1 - f)a + fb$. Direct differentiation gives $a' = -a^2/2$ and $b' = -b^2/2$. Also

$$\Phi^2 = (1 - f)a^2 + fb^2 - f(1 - f)(b - a)^2, \qquad |L_\varepsilon \nabla \Phi|^2 = (b - a)^2 f(1 - f).$$

Adding half of these two identities to $\partial_t \Phi = -((1 - f)a^2 + fb^2)/2$ proves $\partial_t \Phi + |L_\varepsilon \nabla \Phi|^2/2 + \Phi^2/2 = 0$. Its endpoint pairing is $\Phi(1, x_1) - \Phi(0, x_0) = b(1) - a(0) = 4/(1 + \delta)$. Fenchel inequality and (1), followed by $\delta \downarrow 0$, give the lower bound 4. The admissible zero-flux path $G_t = (1 - t)^2\delta_{x_0} + t^2\delta_{x_1}$, $R_t = -2(1 - t)\delta_{x_0} + 2t\delta_{x_1}$ has action 4. Indeed, its reaction is the time derivative of its mass, which verifies (1). Using the reference measure $dt \otimes (\delta_{x_0} + \delta_{x_1})$, each atom with scalar mass $m > 0$ and reaction z has support cost

$$\sup_c \{zc - mc^2/2\} = z^2/(2m),$$

because the zero flux makes the transport dual zero optimal, the mass constraint is saturated, and the maximum occurs at $c = z/m$. Thus the cost is $[-2(1 - t)]^2/[2(1 - t)^2] = 2$ at the first atom for $t < 1$ and $(2t)^2/(2t^2) = 2$ at the second for $t > 0$. At the two exceptional endpoint times the measures have zero mass, so integrating gives $2 + 2 = 4$. At the singular target, the horizontal moving unit atom of velocity $(1, 0)$ has zero reaction and action at most

1/2. Explicitly, push ds forward by $s \mapsto (s, (-1/2 + s, 0))$ to form G and take $q = (1, 0)G$, $R = 0$. The chain rule and interval fundamental theorem give (1) for every closed-domain test. For a scalar admissible dual triple, $A + B_1 \le -B_1^2/2 + B_1 \le 1/2$, which proves the action bound. □

Proposition 7.3 (Frobenius divergence does not ensure a value limit)**.** *For scalar ordinary divergence on the unit disk, $H = 1$, and endpoints $\delta_{(0,-1)}$, $\delta_{(0,1)}$, an alternating sequence $L_{2j} = \operatorname{diag}(t_{2j}, 1)$, $L_{2j+1} = \operatorname{diag}(1, t_{2j+1})$, $0 < t_j \to \infty$, has $\|L_j\|_F^2 \to \infty$ but nonconvergent minimum values.*

Proof. For the even sequence $\Phi(t, x, y) = y/4 - t/8$ satisfies $|\Phi| \le 3/8$ on the time–disk cylinder and $L_{2j}\nabla\Phi = (0, 1/4)$, so it has

$$\partial_t\Phi + \tfrac{1}{2}|L_{2j}\nabla\Phi|^2 + \tfrac{1}{2}\Phi^2 \le -\tfrac{1}{8} + \tfrac{1}{32} + \tfrac{9}{128} = -\tfrac{3}{128}.$$

Its endpoint pairing is $\Phi(1, 0, 1) - \Phi(0, 0, -1) = 1/8 + 1/4 = 3/8$, so the even minima are at least $3/8$ by (1) and the support inequality. For the odd sequence use $\gamma(s) = (0, 2s - 1)$, $G = (s \mapsto (s, \gamma(s)))_{\#}ds$, $q = (0, 2)G$, $R = 0$. The trajectory lies in the interior for interior times, so the chain rule and interval fundamental theorem verify every original weak test. The support constraint gives $A + 2B_2 \le -t_{2j+1}^2 B_2^2/2 + 2B_2 \le 2/t_{2j+1}^2$ eventually. Hence the balanced action is at most $2/t_{2j+1}^2 \to 0$. Thus the values cannot converge. □

Proposition 7.4 (The operator survives uniformly cheap transport)**.** *On the unit disk, for the Kantorovich–Bures operator $D^*\Phi = \operatorname{div}\Phi$, $n = d = 2$, $k = 1$, $H = I_2$, and endpoints $I_2\delta_{(0,-1)}$, $I_2\delta_{(0,1)}$, one has $V(tI_1, I_2) \ge 1/4$ for all $t > 0$, although the total endpoint matrices coincide.*

Proof. Use $\Phi(s, x, y) = \operatorname{diag}(y/4 - s/8, -s/8)$. Its row divergence is $(\partial_x\Phi_{11} + \partial_y\Phi_{12}, \partial_x\Phi_{21} + \partial_y\Phi_{22}) = (0, 0)$. Here $s \in [0, 1]$ is path time. On the unit disk $|y| \le 1$, hence $|y/4 - s/8| \le 3/8$ and $|s/8| \le 1/8$. The diagonal entries of $\partial_s\Phi + \Phi^2/2$ are therefore at most $-1/8 + 9/128 \le -3/128$ and $-1/8 + 1/128 = -15/128$. At the terminal point $(0, 1)$ the trace of $\Phi(1)$ is zero; at the initial point $(0, -1)$ the trace of $\Phi(0)$ is $-1/4$. The endpoint pairing is therefore $1/4$. Thus the dual lower bound is independent of the scalar transport weight. The Bures expression of the total matrices is zero, so the entrywise-divergence identification cannot extend to this operator. □

Remark 7.5. Unscaled flux need not be precompact even for vanishing-gap paths in the large-transport regime. On a unit spatial interval, let $s \in [0, 1]$ denote path time and $\tau > 0$ the transport scale. Take equal endpoints $I_2\,dx$, $H = I_2$, and $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. For entrywise divergence the space–time measures

$$G^{(\tau)} = I_2\,ds\,dx, \qquad q^{(\tau)} = \sqrt{\tau}J\,ds\,dx, \qquad R^{(\tau)} = 0$$

have densities constant in path time. The skew matrix J pairs to zero with every symmetric spatial derivative, so these paths satisfy all symmetric weak tests. Their action at weight τI_2 is $\|J\|_F^2/(2\tau)$, while flux variation is $\sqrt{\tau}\,\|J\|_F$. For this exact calculation, saturate the mass-dual constraint at $g = I_2$. With zero reaction, $C = 0$ is optimal, so the support cost is

$$\sup_B\{B \cdot z - \tfrac{1}{2}\tau^2\,\|B\|_F^2\} = \frac{\|z\|_F^2}{2\tau^2},$$

by completing the square, with equality at $B = z/\tau^2$. Inserting $z = \sqrt{\tau}J$ and integrating over the unit time–space rectangle gives the asserted action exactly. The stationary zero-flux path has action zero, so the gaps vanish as $\tau \to \infty$. Pairing the flux with the constant test selecting its $(1, 2)$ entry gives $\sqrt{\tau}$; thus no weak-star convergent subsequence exists along $\tau \to \infty$. This explains why Theorem 5.3 asserts energy convergence without flux compactness.

8 A smooth positive separation of two models

On the two-dimensional unit disk let $n = d = 2$ and compare the operators

$$D^*_{\mathrm{KB}}\Phi = \operatorname{div}\Phi, \qquad D^*_{\mathrm{full}}\Phi = (\partial_1\Phi, \partial_2\Phi, S_0\Phi - \Phi S_0, S_1\Phi - \Phi S_1),$$

where $S_0 = \operatorname{diag}(1,-1)$ and $S_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. The KB output is a 2×1 column, so $k = 1$. The full output is the horizontal concatenation of four 2×2 blocks, so $k = 8$. These are concrete instances of the models in [7, Examples 7.1 and 7.3]. Their comparison is part of the question posed in [7, Remark 7.1]; Theorem 8.2 gives a separation using smooth, uniformly positive endpoint densities. Their unit-weight minimum actions are

$$V_{\mathrm{KB}}(G_0, G_1) := V_{D^*_{\mathrm{KB}}}(I_1, I_2; G_0, G_1),$$
$$V_{\mathrm{full}}(G_0, G_1) := V_{D^*_{\mathrm{full}}}(I_8, I_2; G_0, G_1).$$

The original Kantorovich–Bures normalization is $d^2_{\mathrm{KB}} := 2V_{\mathrm{KB}}$ [2, Definition 2.1 and equation (2.3)]. On spatially constant symmetric fields, the KB operator vanishes, while the full operator vanishes precisely on the common commutant of S_0, S_1.

Lemma 8.1. *The common commutant of S_0, S_1 consists precisely of scalar matrices.*

Proof. For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, $S_0A - AS_0 = \left(\begin{smallmatrix} 0 & 2b \\ -2c & 0 \end{smallmatrix}\right)$, so commutation with S_0 forces $b = c = 0$. For such a diagonal matrix, $S_1A - AS_1 = \left(\begin{smallmatrix} 0 & d-a \\ a-d & 0 \end{smallmatrix}\right)$; commutation with S_1 therefore forces $a = d$. Conversely, scalar matrices commute with both generators. □

Theorem 8.2 (Smooth uniformly positive separation)**.** *There exist smooth matrix densities G_j on the disk and $c > 0$ such that*

$$\tfrac{1}{2}I_2 \preceq G_j \preceq \tfrac{3}{2}I_2, \qquad G_j\,dx \xrightarrow{*} I_2\,dx, \qquad V_{\mathrm{full}}(I_2\,dx, G_j\,dx) \to 0,$$

while $c \le V_{\mathrm{KB}}(I_2\,dx, G_j\,dx) \le \pi/4$ for every j. There is no tail-uniform finite constant C with $V_{\mathrm{KB}}(I_2\,dx, G_j\,dx) \le C V_{\mathrm{full}}(I_2\,dx, G_j\,dx)$.

Proof. Here G_j denotes a spatial density, and $G_j\,dx$ its endpoint measure. Choose $\chi \in C^\infty_c(B_{1/2})$, $0 \le \chi \le 1$, $\chi(0) = 1$. Let $\kappa_j = 2\pi(j+1)$, $E = \operatorname{diag}(0,1)$ and

$$h_j(x) = \frac{\chi(x)\sin(\kappa_j x_1)}{\kappa_j}, \quad f_j = \partial_1 h_j, \qquad G_j = I_2 + \tfrac{1}{4}f_j E.$$

Expanding the derivative gives

$$f_j = \chi\cos(\kappa_j x_1) + \kappa_j^{-1}(\partial_1\chi)\sin(\kappa_j x_1).$$

Choose a fixed integer shift so that $\kappa_j \ge \|\partial_1\chi\|_\infty$ at every relabeled index. Then $|f_j| \le 2$, and the two diagonal entries of G_j lie between $1/2$ and $3/2$. Compact support gives $\int f_j = \int \partial_1 h_j = 0$ by one-dimensional integration in x_1 followed by Fubini.

We spell out the oscillatory limit once. For a continuous scalar test ψ, extend $a = \chi\psi$ by zero outside the disk. This is continuous with compact support, hence integrable. The Riemann–Lebesgue lemma gives the required decay; its mechanism here is a half-period translation. For $k > 0$ set $v_k = (\pi/k)e_1$ and $I_k = \int a(x)\cos(kx_1)\,dx$. Translation invariance of Lebesgue measure and the sign change of the cosine give

$$I_k = \frac{1}{2}\int \big(a(x) - a(x+v_k)\big)\cos(kx_1)\,dx.$$

For large k the difference is supported in a fixed compact set. Uniform continuity makes its supremum norm tend to zero as $v_k \to 0$, and the integral therefore tends to zero. This proves the vanishing of the leading oscillatory term against every continuous test. The derivative term in f_j tends uniformly to zero, so each coordinate of $G_j\,dx$ converges weak-star to the corresponding coordinate of $I_2\,dx$.

The small antiderivative h_j supplies a low-action full-model flux; the matrix direction E also permits a KB dual field with zero row divergence. For the full operator use the mass density $M(s,x) = I_2 + sf_j(x)E/4$, first spatial block $Q_1 = -h_jE/4$, and every other flux and reaction block zero. Here $\dot{M} = f_jE/4 = -\partial_1 Q_1$. Integration by parts in time produces $-\int \Phi \cdot \dot{M}$ after the endpoint terms; integration by parts in space produces $-\int \Phi \cdot \partial_1 Q_1$, which cancels it. The spatial flux is compactly supported in $B_{1/2}$, so no boundary term occurs, for any closed-domain test. This verifies (1). For any admissible dual triple at unit weights, pairing its constraint with $M \succeq I_2/2$ and dropping the other nonnegative channel penalties gives

$$A \cdot M + B_1 \cdot Q_1 \le -\tfrac{1}{2}\operatorname{tr}(B_1 B_1^T M) + B_1 \cdot Q_1 \le -\tfrac{1}{4}\|B_1\|_F^2 + B_1 \cdot Q_1 \le \|Q_1\|_F^2 = \tfrac{1}{16}h_j^2.$$

The last inequality follows by subtracting $\|B_1 - 2Q_1\|_F^2/4$ from $\|Q_1\|_F^2$. Taking the support supremum and integrating over unit time bounds its action by

$$\tfrac{1}{16}\int h_j^2 dx \le \frac{\|\chi\|_{L^2}^2}{16\kappa_j^2} \to 0.$$

For the KB lower bound, set $b_j = \int \cos(\kappa_j x_1)E \cdot (G_j - I_2)dx$. Integration by parts gives

$$b_j = \tfrac{1}{4}\int \cos(\kappa_j x_1)\partial_1 h_j\,dx = \tfrac{1}{4}\int \chi \sin^2(\kappa_j x_1)dx \longrightarrow \tfrac{1}{8}\int \chi dx > 0.$$

For the second equality, integration by parts differentiates the cosine into $-\kappa_j \sin(\kappa_j x_1)$ and cancels the factor $1/\kappa_j$ in h_j. For the limit, use $\sin^2 z = (1 - \cos(2z))/2$ and the preceding oscillatory argument with $\psi = 1$. Its limit is positive because $\chi(0) = 1$ and χ is continuous and nonnegative. Also $0 \le b_j \le \pi/4$, so $a_j = b_j/(2\pi)$ lies in $[0, 1/8]$. The KB test $\Phi_j(s,x) = (a_j \cos(\kappa_j x_1) - a_j^2 s)E$ has row divergence zero: its only nonzero entry is the $(2,2)$ entry, and this entry is independent of x_2. Also

$$\partial_s \Phi_j + \tfrac{1}{2}\Phi_j^2 \preceq a_j^2(-1 + \tfrac{1}{2}(1 + a_j)^2)E \preceq 0.$$

Indeed, $|a_j \cos(\kappa_j x_1) - a_j^2 s| \le a_j(1 + a_j)$ and $(1 + a_j)^2/2 \le 81/128 < 1$, which proves the displayed dual inequality. The identity $\int f_j = 0$ gives $\int E \cdot G_j\,dx = \pi$, so its endpoint pairing is $a_j b_j - a_j^2\pi = b_j^2/(4\pi)$. If $b_\infty = (1/8)\int \chi > 0$, a further fixed shift ensures $b_j \ge b_\infty/2$ for every relabeled index. Thus one may take $c = b_\infty^2/(16\pi) > 0$ in the lower bound.

For the KB upper bound put $u_j = f_j/4$ and use the zero-flux pure-reaction curve

$$M(s,x) = \operatorname{diag}(1, (1 + s(\sqrt{1 + u_j(x)} - 1))^2).$$

Its reaction is its time derivative, so time integration by parts verifies admissibility. Put $a(x) = \sqrt{1 + u_j(x)} - 1$. The moving diagonal entry $m = (1 + sa)^2$ is positive, because $1 + sa$ lies between 1 and $\sqrt{1 + u_j} \ge 1/\sqrt{2}$, and its reaction entry is $z = 2a(1 + sa)$. As in the scalar calculation in Proposition 7.2, maximizing the mass dual at zero flux leaves $C \cdot \dot{M} - \operatorname{tr}(CC^T M)/2$. Since $M = \operatorname{diag}(1, m)$ and $\dot{M} = \operatorname{diag}(0, z)$, this equals

$$zC_{22} - \tfrac{1}{2}(C_{11}^2 + C_{12}^2) - \tfrac{m}{2}(C_{21}^2 + C_{22}^2).$$

Setting the other entries to zero and $C_{22} = z/m$ shows that its supremum is exactly $z^2/(2m) = 2a^2$. The action is consequently $2\int(\sqrt{1+u_j}-1)^2dx$. For $|u_j| \le 1/2$,

$$2(\sqrt{1+u_j}-1)^2 \le 2(1-1/\sqrt{2})^2 = 3-2\sqrt{2} < \tfrac{1}{4},$$

so integration over the disk gives the required upper bound $\pi/4$. Finally a fixed comparison with $V_{\mathrm{full}} \to 0$ would force the positive KB lower bound to zero, a contradiction. □

9 Numerical illustrations

We compare the exact rotating-reaction values with numerical approximations to finite-dimensional SDP minima on the unit interval, with $n = 2$, $d = 1$, and entrywise transport dimension $k = 2$. Let N_t, N_x be positive integers and set $\Delta s = 1/N_t$, $\Delta x = 1/N_x$. Mass matrices $M_{k,i} \in \mathbb{S}^2_+$ are sampled at time nodes $k = 0, \dots, N_t$ and cell centres $i = 1, \dots, N_x$. Here k is a local time index. Full flux matrices $Q_{k,i+1/2} \in \mathbb{R}^{2\times 2}$ are sampled at time midpoints $k = 0, \dots, N_t - 1$ and interior faces $i = 1, \dots, N_x - 1$; full reactions $R_{k,i} \in \mathbb{R}^{2\times 2}$ use the same time midpoints and all cells. Exterior fluxes satisfy $Q_{k,1/2} = Q_{k,N_x+1/2} = 0$. For $0 \le k < N_t$ and $1 \le i \le N_x$, the discrete continuity equation is

$$\frac{M_{k+1,i} - M_{k,i}}{\Delta s} + \frac{\operatorname{sym} Q_{k,i+1/2} - \operatorname{sym} Q_{k,i-1/2}}{\Delta x} = \operatorname{sym} R_{k,i}.$$

The endpoint matrices are fixed and every nodal mass is PSD. The mass used in each reaction cost is the arithmetic mean of its two neighbouring time samples; the face mass additionally averages the two adjacent cells. For each channel, we impose the 4×4 PSD constraint

$$\begin{pmatrix} \overline{M} & Z \\ Z^T & S \end{pmatrix} \succeq 0, \qquad Z = QL^{-1} \text{ or } RH^{-1}, \qquad S \in \mathbb{S}^2,$$

where Q is the corresponding local flux matrix and each channel has its own auxiliary matrix S. The objective is $\Delta s \Delta x/2$ times the sum of the auxiliary traces, with the same factor $1/2$ as (3). All weights passed to the solver are positive definite; the balanced comparison instead sets $R = 0$. We use CVXPY 1.7.2 with Clarabel 0.11.1, absolute and relative gap and feasibility tolerances 10^{-9}, and a limit of 300 iterations. Postprocessing uses a threshold of 10^{-7} for absolute and scaled continuity-equation and endpoint residuals, scaled cone violations, symmetry errors, boundary fluxes and range residuals; mass eigenvalues must be at least -10^{-7}. The epigraph action and independently recomputed quadratic action J_{quad} must differ by at most $10^{-7} + 10^{-6}|J_{\mathrm{quad}}|$.

Figure 1(a) evaluates the exact formula in Proposition 7.1 along its three reaction regimes $\theta = \varepsilon^{3/4}, \varepsilon^{1/2}, \varepsilon^{1/4}$, whose limits as $\varepsilon \to 0$ are V_*, $V_* + 1/2$ and $+\infty$. For panel (b), $N_x = 1$ and $N_t = 8, 16, 32$; the two reaction weights have $(\varepsilon, \theta) = (0.4, 0.2)$ and $(0.2, 0.35)$. The absolute differences from the continuum formula decrease by about a factor of four at each temporal refinement, from approximately 1.65×10^{-4} and 1.69×10^{-4} to 1.03×10^{-5} and 1.06×10^{-5}, respectively.

For panel (c), take $N_x = 4$, $N_t = 16$, $H = I_2$, $L = tI_2$ and sample

$$G_0(x) = I_2 + \tfrac{1}{4}\cos(2\pi x)\operatorname{diag}(1,-1), \qquad G_1(x) = I_2 + \tfrac{1}{4}\sin(2\pi x)\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

at the cell centres. Both continuum total matrices equal I_2, so $E_H = 0$. Their traces are identically two, and their hub moments about $x_* = 1/2$ are $2\int_0^1 (x-1/2)^2 dx = 1/6$. Thus Theorem 6.2 gives the continuum upper bound $1/(3t^2)$. For $t = 1, 2, 4$, the computed grid

values are approximately 0.003604, 0.0009564 and 0.0002429. On the same grid with $L = I_2$, reaction weights $H = \varepsilon I_2$, $\varepsilon = 1, 1/4, 1/16$, give values increasing from 0.003604 to 0.003905, towards the balanced grid value 0.003906.

All plotted solver outputs pass the recorded feasibility and objective checks; the largest recorded primal–dual gap is below 4.5×10^{-10}. These checks include solver statuses, residuals and independent quadratic-cost evaluations. Panels (b) and (c) compare the stated continuum references with the SDP values on the specified grids.

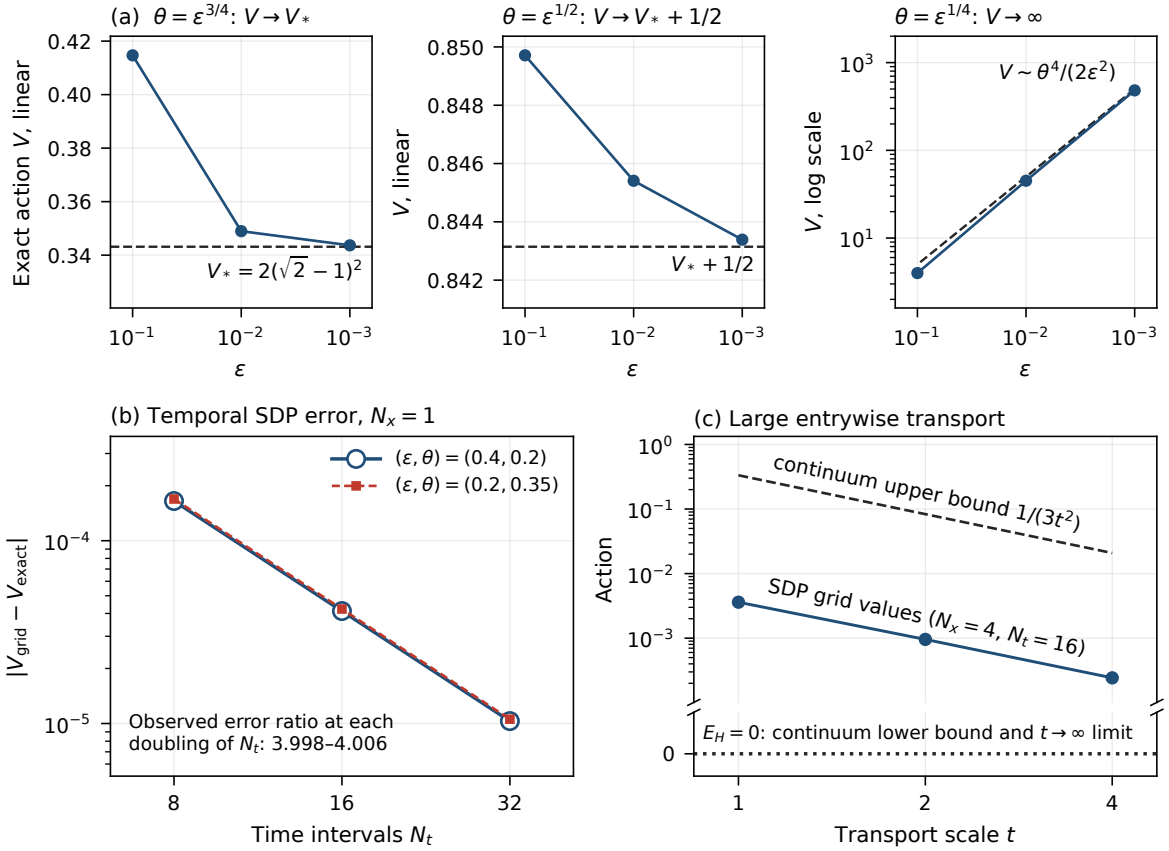


Figure 1: (a) Upper three facets: exact minimum action V of Proposition 7.1 for $\theta = \varepsilon^{3/4}, \varepsilon^{1/2}, \varepsilon^{1/4}$ at $\varepsilon = 10^{-1}, 10^{-2}, 10^{-3}$, on separate vertical axes (linear, linear, logarithmic). Dashed lines mark the limits $V_* = 2(\sqrt{2}-1)^2$ and $V_* + 1/2$ and the asymptotic reference $V \sim \theta^4/(2\varepsilon^2)$ of the same proposition. (b) Absolute differences between SDP values with $N_x = 1$ and exact values, on common axes, for $(\varepsilon, \theta) = (0.4, 0.2)$ (circles, solid line) and $(0.2, 0.35)$ (squares, dashed line). (c) SDP values on the fixed grid $N_x = 4$, $N_t = 16$ (circles) and the continuum upper bound $1/(3t^2)$ of Theorem 6.2 (dashed). The separate linear strip below the axis break marks $E_H = 0$ (dotted), the continuum lower bound and the $t \to \infty$ limit.

10 Formal verification and reproducibility

The ancillary materials contain Lean formalizations of Theorems 4.1, 5.2, 5.3, 6.2 and 8.2, Propositions 7.1–7.4, and the supporting path constructions. They use signed Radon measures, the complete closed-domain C^1 test class and the normalization in (3). The accompanying source inventory and verification manifests identify the proof files by their hashes. The pinned toolchain is Lean 4.34.0-rc2; its mathlib revision and the checker versions are specified in the

accompanying manifests. The recorded kernel certificate contains the Lean trust-level-zero check and the independent Nanoda check of the exported declarations and their complete dependency closures, with `propext`, `Classical.choice` and `Quot.sound` as the permitted axioms.

The ancillary statement map connects the manuscript results to exact declaration types and source hashes. The numerical code, figure inputs, recorded outputs and reconstruction instructions are included in the same ancillary directory.

11 Conclusion

In the signed-Radon framework of Li and Zou [7], Theorem 4.1 gives stability of extended minimum values for convergent weights under eventual relative Gram comparison with factors tending to one. At finite target cost, the minimum is attained, near-minimizers admit optimal weak-star clusters, and Frobenius variations vanish in the null directions of the limiting weights. The fixed-basis, reaction-zero and partial-reaction results are specializations of this comparison mechanism.

For large transport scalings with fixed reaction weight, the limiting reaction-channel infimum retains the original finite-baseline class. Theorems 5.2 and 5.3 treat fixed, possibly singular rays and uniform divergence in every direction separately. For entrywise divergence on a convex domain with positive-definite reaction weight, the hub construction identifies the latter limit with the weighted Bures value of the total endpoint matrices and gives an explicit recovery bound.

The rotating families and norm-divergent sequence exhibit failures of value continuity and of magnitude-based convergence assumptions. The unscaled-flux example separates convergence of actions from compactness of fluxes. The Kantorovich–Bures obstruction and smooth positive separation show that the operator can affect both large-weight limits and costs along a common weak-star convergent family of uniformly positive densities.

AI declaration

Generative AI tools contributed to the mathematical development and the organization and revision of this manuscript. The author provided initial ideas, made substantive contributions to refining the results, reviewed the arguments, and takes full responsibility for the final content.

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